

STAT 5309

R LAB 1

****CONTENTS: 2 SAMPLE T-TESTS (EQUAL/UNEQUAL VARIANCES/PAIRED)**

***DUE: Thurs, Jan 31**

A. PRACTICE

----- SET UP DATA-----

Data: Tension Bond Strength.

One factor of 2 levels: Modified mortar and Unmodified mortar

Modified	Unmodified
16.85	16.62
16.40	16.75
17.21	17.37
16.35	17.12
16.52	16.98
17.04	16.87
16.96	17.34
17.15	17.02
16.59	17.08
16.57	17.27

```
y1 <- c(16.85,16.40,17.21,16.35,16.52,17.04,16.96,17.15,16.59,16.57)
```

```
y2 <- c(16.62,16.75,17.37,17.12,16.98,16.87,17.34,17.02,17.08,17.27)
```

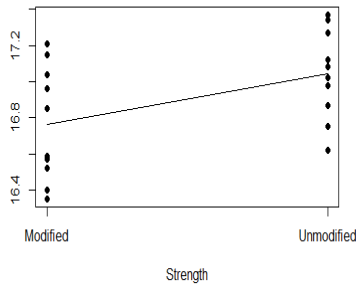
```
y <- data.frame(y1,y2)
```

```
names(y) <-c("Modified", "Unmodified") #change column names
```

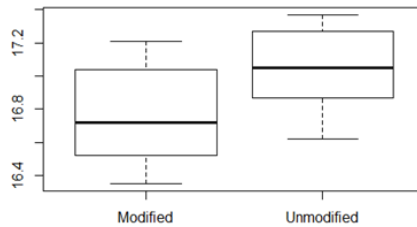
```
y.means <- apply(y,2,mean) #2 for column means, 1 for row means
```

#-----stripchart(); boxplot()-----

```
stripchart(y, xlab="Strength", vertical=TRUE,pch=16) #Stripchart  
lines(y.means)
```



boxplot(y)



#boxplot

```
##-----TWO-SAMPLE T-TEST (Unknown, Equal/Unequal Variance, )-----
##-----PAIRED T-TEST -----
## -----OTHER TESTS
```

```
> t.test(mod,unmod,var.equal=TRUE) # with unknown equal variance t-test,
```

Two Sample t-test

data: mod and unmod

t = -9.1094, **df** = 18, p-value = 3.678e-08

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-1.4250734 -0.8909266

sample estimates:

mean of x mean of y

16.764 17.922

```
> t.test(mod,unmod, var.equal=FALSE)
```

Welch Two Sample t-test

data: mod and unmod

t = -9.1094, df = 17.025, p-value = 5.894e-08

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-1.4261741 -0.8898259

sample estimates:

mean of x mean of y

16.764 17.922

#-----**Paired t-test**-----

Data: Hardness Test experiment.

A rod with a pointed tip is pushed into a metal specimen with a known force. 2 different tips are used, by measuring the depression caused by the tip, the hardness of the metal is determined. Metal specimens(20) .Half of these is tested by tip 1 and half by tip 2.

Specimen	Tip 1	Tip 2
	7, 3, 3, 4, 8, 3, 2, 9, 5, 4	6, 3, 5, 3, 8, 2, 4, 9, 4, 5

```
>y1<- c(7, 3, 3, 4, 8, 3, 2, 9, 5, 4)
```

```
>y2 <-c(6, 3, 5, 3, 8, 2, 4, 9, 4, 5)
```

```
>harness<- data.frame(y1, y2)
```

```
>t.test( y1, y2,paired=T)
```

#-----**Test on the Differences (against 0, using t.test())**

```
>diff <- y2-y1
```

```
>t.test(diff)      #should have same output as above
```

##-----**Wilcox.test() : Nonparametric test**-----

```
wilcox.test(y1,y2)
```

```
wilcox.test(y~tip,data=harness,paired=TRUE)
```

##-----**Variances test: test whether variances are equal**-----

```
>var.test(mod,unmod)
```

F test to compare two variances

data: mod and unmod

F = 1.6293, num df = 9, denom df = 9, p-value = 0.4785 alternative hypothesis: true ratio of variances is not equal to 1 95 percent confidence interval: 0.4046845 6.5593806 sample estimates: ratio of variances 1.629257	# variances are equal
---	-----------------------

B. EXERCISE

1. Data: **Bottle Filling Machines.** Two machines are used to fill plastic bottles with a volume of 16.0 ounces. The process can be assumed to be normal, with $\sigma_1 = 0.015$, $\sigma_2 = 0.018$. An experiment is performed

Machine 1	Machine 2
16.03 16.01 16.04 15.96 16.05	16.02 16.03 15.97 16.04 15.96
15.98 16.05 16.02 16.02 15.99	16.02 16.01 16.01 15.99 16.00

Perform by R (or by calculator to check).

- (a) Test hypothesis: $H_0: \mu_1 = \mu_2$. What test do you use? Perform the test.
- (b) Find the P-value for the test.
- (c) Find the 95%-CI for the difference of the population means.
- (d) Perform a boxplot and a stripchart with line connecting means.

2. Data: **Burning times of Chemical Flares.**

Type 1	Type 2
65 82 81 67 57	64 56 71 69 83
59 66 75 82 70	74 59 82 65 79

Perform by R . Only use var.test() to check

- (a) Test the hypothesis that the two variances are equal. $\alpha = .05$
- (b) Test the hypothesis that the burning times are equal. Test statistics. P-value. Conclusion.
- (c) Check the normality assumption for both types of flares.